AE 481 Team 01 Assignment 6

October 9, 2025

1 Design Point

A new design point was selected using the T/W-W/S plot. An initial planform analysis found that the aircraft length constraint of 50 ft from the RFP posed a challenge for the relatively large wings. However, increasing wingspan allowed an aircraft to maintain a similar performance (i.e. combat radius, dash speed), while reducing wing area. Therefore, the design point was selected using the largest wing span possible of 60 ft.

Additionally, the catapult takeoff requirements were found to consistently be the most constraining sizing constraint. To prevent this from limiting aircraft performance, an aircraft $C_{L,\max}$ of 2.0 was selected. This was made based on assumptions that the aircraft would utilize slotted flaperons for takeoff. The $C_{L,\max}$ was determined using fits from Raymer [1].

The design point selected has a thrust-to-weight ratio (TWR) of 0.78 and a wing loading (W/S) of 114 lb/ft^2. For the air-to-air mission, the combat radius is 900 nm with a dash Mach number of 1.7 and a combat time of 2.5 min. For the strike mission, the combat radius is 1020 nm with a sea level dash Mach number of 0.7.

The planform area for the design point is 670 ft^2. This was selected since it was the lowest wing area that maintained high mission performance such as combat range. Additionally, it was judged to be small enough within the RFP dimension constraints. The sizing plots for this design point are shown in Figure 1 and Figure 2. The current design point exceeds the RFP performance requirements, but results in a large aircraft. This potentially introduces issues with the aircraft height, length, and stowed spot factor. Further analysis will be conducted in the future to revise the design point, such as performing proper trades on mission parameters with aircraft geometry parameters.

2 Lifting Surfaces Layout

A picture of the overall layout is shown in Figure 3. The selection of the dimensions is discussed in the following sections.

2.1 Wing Layout

The wing was parameterized in terms of the taper ratio (λ), sweep angle ($\Delta_{c/4}$), and root-to-tip twist angle (θ_{twist}). Root incidence angle was set to zero. The aircraft wingspan was maximized to the RFP requirement of 60 ft, and the wing area was kept constant using the results from the sizing. This results in varying aspect ratio (AR). The coefficient of lift was calculated from the cruise segment start weight of 71,893 lb, which

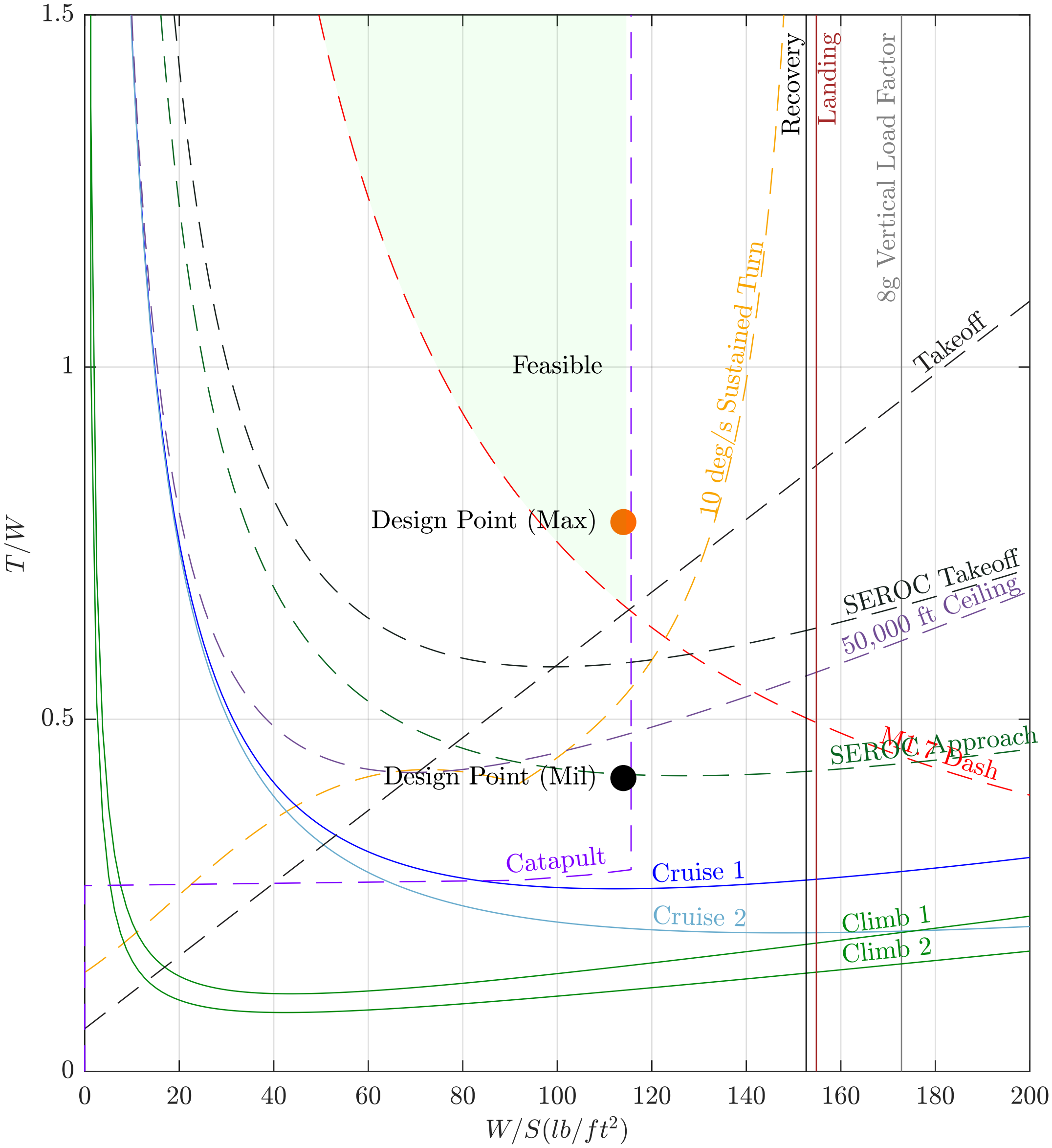

Figure 1: T/W-W/S sizing plot for the air to air mission. Dashed lines represent maximum thrust constraints. Solid lines are military thrust.

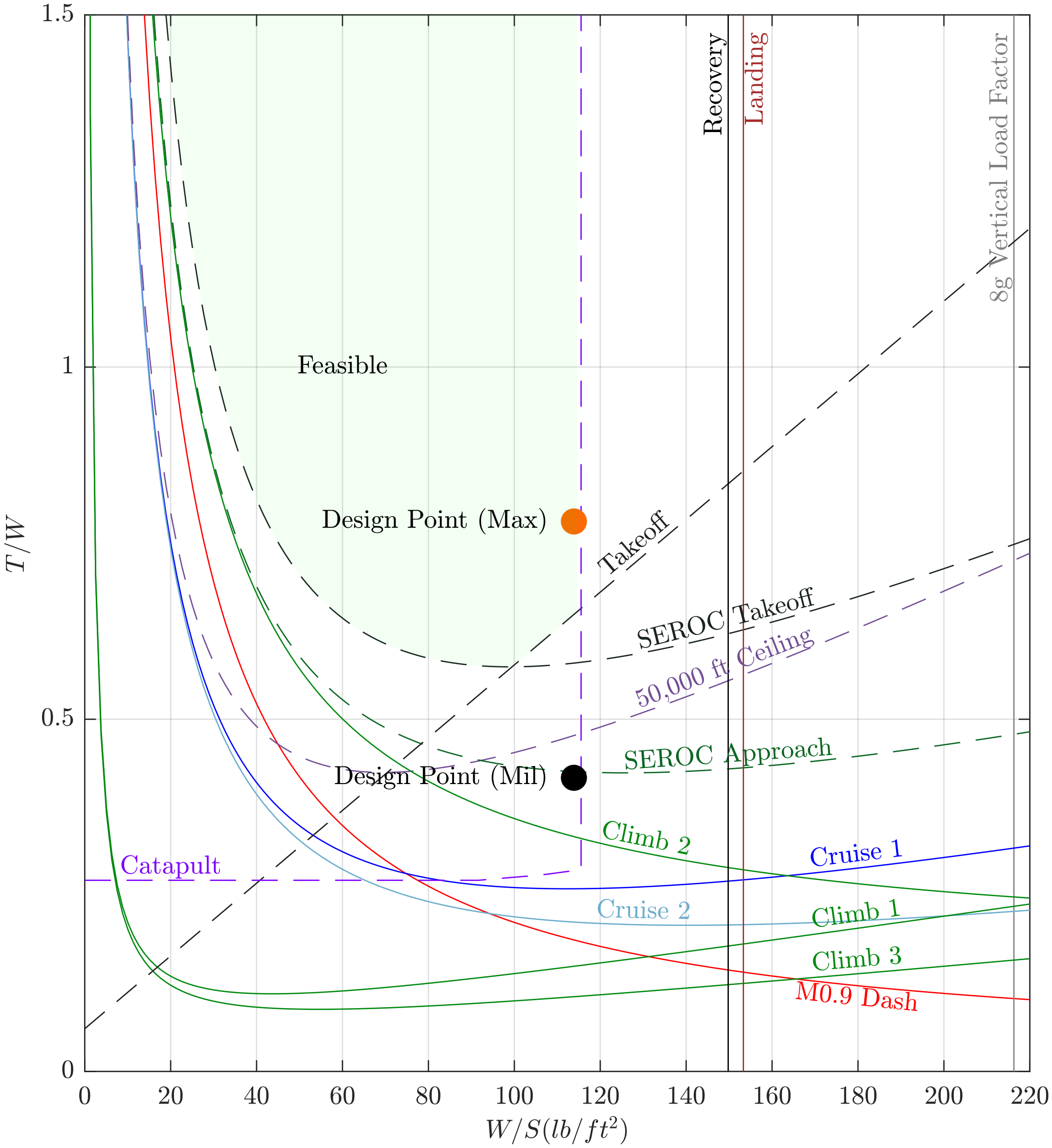

Figure 2: T/W-W/S sizing plot for the strike mission. Dashed lines represent maximum thrust constraints. Solid lines are military thrust.

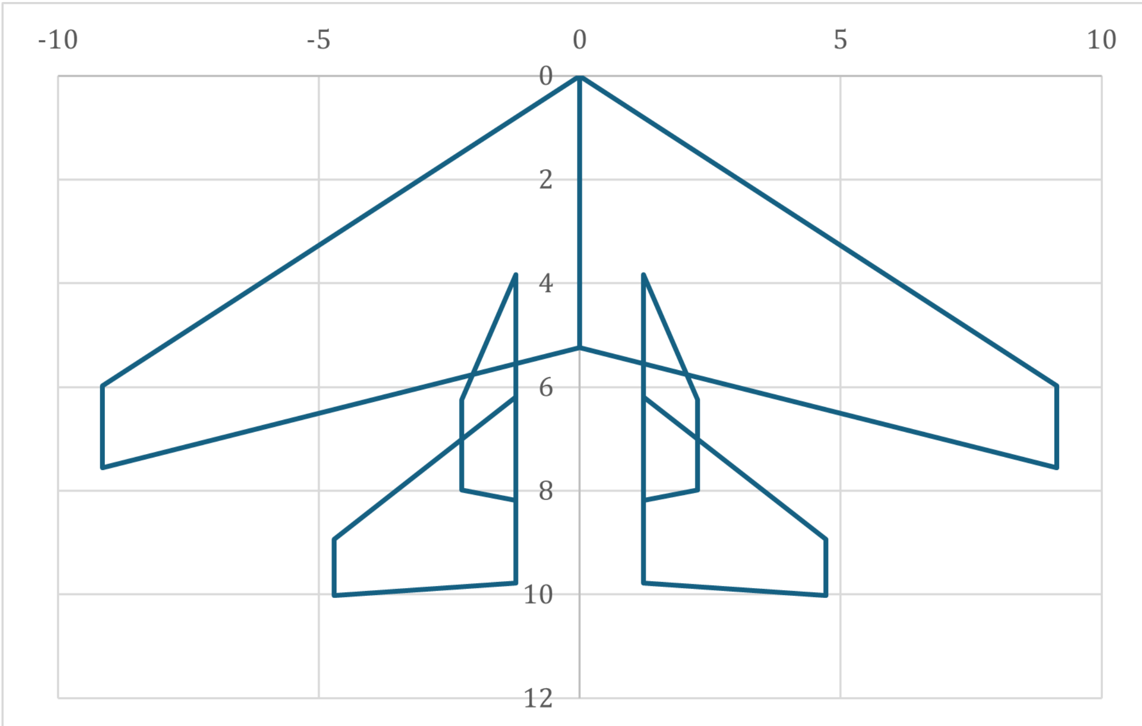

Figure 3: Diagram of the Layout. Dimensions in meters.

results in a C_L of 0.555. The objective function for the optimization was to maximize L/D at the cruise condition, or alternatively minimize total drag.

For modeling of the 3D aerodynamic characteristics, AVL (Athena Vortex Lattice) [2] was employed. The drag components considered were the induced drag C_{Di}, wing parasitic drag C_{Dp}, and wing wave drag C_{Dw}. A constant $C_{D,misc}$ of 0.01 was added to account for the fuselage, tails, and interferences. C_{Di} was computed in AVL, and read from the output CDff. C_{Dp} was factored in using the drag polar of our airfoil. This was input to AVL using the CDCL keyword, and read through the output CDvisc. The CDCL points were tailored to achieve greatest accuracy where C_ℓ is between 0 and 0.6, matching the wing C_ℓ distribution.

The C_{Dw} of the wing was calculated using the process discussed in Section 6.4 of the Metabook [3], assuming our airfoil can achieve a technology factor of 0.95. For the trades using RANS polars, the C_{Dw} of wing with $\Delta_{c/4} = 0$ was also subtracted, as ADflow models shocks. The swept C_{Dw} is still added, which results in correct behavior as the $\Delta_{c/4}$ increases.

An Excel tool was written that automatically interfaces with AVL using Visual Basic. The Excel script calculates the relevant dimensions for a user selected combination of λ, $\Delta_{c/4}$, and θ_{twist}, which are input into a .txt file for AVL to read. 20 chordwise and 10 spanwise panels were used, which results in a high accuracy and short computational time [2]. Trades were conducted manually using the Excel script, and data was recorded and processed in a separate sheet.

The wing trade study was conducted by varying λ in increments of 0.01, and $\Delta_{c/4}$ in increments of 1 degree. For each combination, the θ_{twist} was varied in increments of 0.1 degree, and set to maximize L/D. Trial and error was used to find a range containing close to the best design. The results of the trade study are plotted in Figure 4. The design that maximized the L/D ratio has a λ of 0.3, a $\Delta_{c/4}$ of 29 degrees, and a θ_{twist} -2.7 degrees. This results in a Δ_{LE} of 33.21 degrees.

Taper Ratio	Sweep 26	27	28	29	30	31
0.27	13.973	13.9793	13.985	13.9861	13.9859	13.985
0.28	13.9807	13.9882	13.9916	13.9928	13.9926	13.9918
0.29	13.9872	13.9935	13.9968	13.9979	13.9977	13.9968
0.3	13.99	13.9963	13.9995	14.0017	14.0006	13.9999
0.31	13.9824	13.9889	13.9924	13.9938	13.9937	13.9929
0.32	13.9835	13.9902	13.9938	13.9952	13.9953	13.9946

Figure 4: Trade Study Results to Maximize L/D. Contours from red to green, showing worst to best design respectively.

2.2 Horizontal Tail Layout

With the wing parameters decided, the horizontal and vertical tails can be laid out. The horizontal tail AR was selected to be 3, and the λ was selected to be 0.4 based on recommendations from Raymer Section 4.5.4. The $\Delta_{c/4}$ was selected to be 31.3 degrees, which results in a Δ_{LE} of 38.22 degrees. This is about 5 degrees greater than the wing leading edge, which is desirable according to Raymer 4.5.4. The tail volume coefficient (c_{HT}) was chosen to be 0.4, and was reduced by 20% resulting in a c_{HT} of 0.32. 10% of this reduction is because the elevator is all moving. The additional reduction is due to the relatively large wing area of this aircraft compared to other fighter aircraft. This value will be refined with additional studies. The moment arm length was selected to be 30% of the fuselage length, which reduced the wing and empennage total length without the surfaces overlapping.

2.3 Vertical Tail Layout

The vertical tail was sized in the same way as the horizontal tail. The AR was selected to be 1, and the λ was selected to be 0.4. The λ was maximized within the recommended range to minimize the root chord, and thus the overlap between the wing and horizontal tail. The $\Delta_{c/4}$ was selected to be 30 degrees, which results in a Δ_{LE} of 38.37 degrees. The c_{VT} was selected to be 0.035, which is lower than historically recommended. However, AVL predicts a $C_{n\beta}$ of 0.1 for this configuration, which is high and positively stable. The reason for the small c_{VT} is likely due to the large wingspan, and this value will be refined in the future. The moment arm was selected to be 15% of the fuselage length, which positions the vertical tail between the wing and the horizontal tail. The moment arm was moved back until the back of the vertical tail root intersects 50% of the horizontal tail root. This allows for the elevator to clear the vertical tail.

3 Control Surfaces and Flaps

Throughout this section, the CAD software Siemens NX was used to sketch and locate the control surfaces.

3.1 Elevator and Rudder

The elevator was selected to be all-moving, which reduces horizontal tail size and increases maneuverability. The rudder average chord fraction was selected to be 30% based on Raymer Section 6.6. The full span of the rudder is used to maximize maneuverability. A constant-width control surface was selected to improve manufacturability. This results in an inboard chord fraction of 21.01%, and an ourboard chord fraction of 52.53% at the tip.

3.2 Ailerons

The ailerons were sized primarily based on Raymer Fig. 6.3. The inboard span fraction was selected to be 56.39%, which is where the hinged portion of the wing starts. The outboard span fraction is 95% of the span, to maximize effectiveness. The last 5% is not necessary due to 3D flow effects at the tip. A constant chord hinge was selected to reduce width at the tip. Raymer Fig. 6.3 was used to select the chord fraction based on historical guidelines, which resulted in a chord fraction of 26%.

3.3 Flaps

For the leading edge flaps, a constant width was chosen to increase manufacturability. An average chord fraction of 15% was selected using Raymer guidelines in Section 6.6. The full usable span was used, which results in a span fraction of 26.66% to 100%. This results in an inboard chord fraction of 10.26% and an outboard chord fraction of 27.82%.

For the trailing edge flaps, a constant width was also chosen. An average chord fraction of 25% was selected. The usable span inboard of the folding location was used, which results in a span fraction of 26.66% to 56.39%. This results in an inboard chord fraction of 19.82% and an outboard chord fraction of 26.63%.

4 Airfoil Design

The airfoil was designed using the MACH-Aero framework [4]. The optimizer starts with an airfoil similar to the one used by the F/A-18E, the NACA 65-206 airfoil [5]. The optimization problem is centered around two operating conditions: cruise and supersonic dash. For cruise, the design of $c_l = 0.55$ is calculated using the weight at the beginning of the first cruise segment. A 5-point stencil is constructed about the cruise operating condition, varying c_l and Mach number. This has the effect on ensuring minimal wave drag at, and around the design cruise condition, and not over optimizing for a single condition. For the supersonic dash condition, $c_l = 0.08$. The drag coeffcient between cruise and dash is weighted equally.

Additional constraints include a minimum thickness ratio between the original airfoil, and a minimum volume or cross-sectional area. The thickness requirement prevents the leading and trailing edge from becoming excessively thin, while the volume constraint is useful for maintaining the fuel capacity and structural properties of the airfoil. For example, reducing area would result in a decrease in the fuel carrying capacity, while a thinner airfoil may have worse bending stiffness in a wing. The optimization problem is outlined below.

$$
\begin{aligned}
\text{minimize} \quad & 0.1c_{d,0} + 0.1c_{d,1} + 0.1c_{d,2} + 0.1c_{d,3} + 0.1c_{d,4} + 0.5c_{d,5} \\
\text{with respect to} \quad & \text{angle of attack, shape variables} \\
\text{subject to} \quad & c_{l,0} = 0.55, M = 0.84, h = 40000 \text{ ft} \\
& c_{l,1} = 0.65, M = 0.84, h = 40000 \text{ ft} \\
& c_{l,2} = 0.45, M = 0.84, h = 40000 \text{ ft} \\
& c_{l,3} = 0.55, M = 0.85, h = 40000 \text{ ft} \\
& c_{l,4} = 0.55, M = 0.83, h = 40000 \text{ ft} \\
& c_{l,5} = 0.08, M = 1.7, h = 30000 \text{ ft} \\
& V \geq V_0 \\
& t \geq 0.2847 t_0
\end{aligned}
$$

The resulting airfoil has a c_d of 0.0084 at the cruise condition and a c_d of 0.0212 at the supersonic dash condition. The optimized airfoil is shown in Figure 5. The airfoil top surface has been flattened to reduce the effect of wave drag. Additionally, the leading edge is much thinner to reduce drag at the supersonic dash condition.

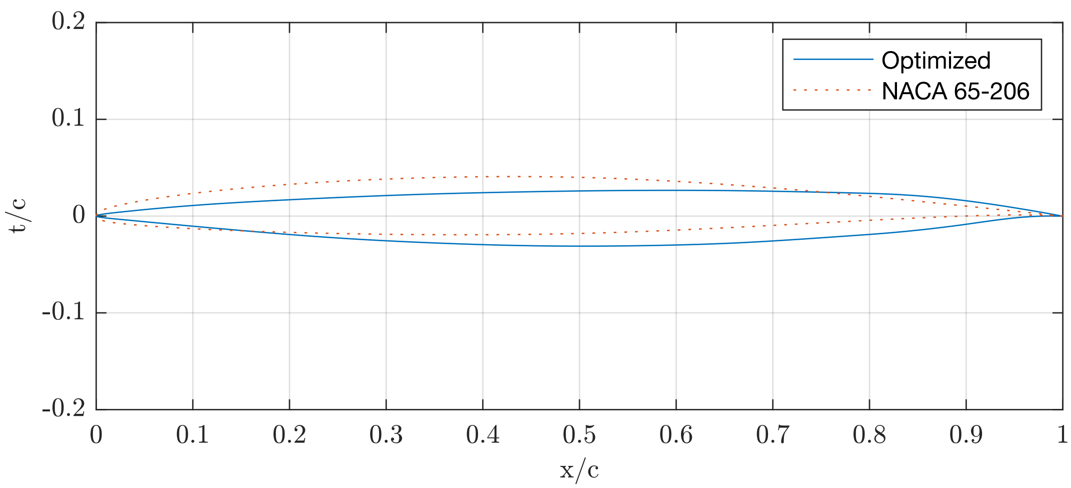

Figure 5: Optimized airfoil and initial NACA 65-206 airfoil.

Plotting the c_p distribution (Fig. 6) for the reveals a flat distribution on the upper surface, and a pressure increase corresponding to a weak shock. This correlates well to the description of a supercritical airfoil in Nicolai [6]. Further analysis on the airfoil will be performed, including drag polars at various Reynolds numbers, to characterize it's impact on the aircraft planform and aerodynamic performance.

4.1 Stabilizer Airfoil

For the horizontal and vertical stabilizer, a biconvex airfoil was selected, based on example sharp-nosed airfoils in Nicolai. This provides a symmetric shape for the stabilizers, and has increased thickness along most of the chord when compared to a double wedge airfoil. The increased thickness can be beneficial for reducing structural weight, and increase internal volume for storing fuel in the vertical stabilizers. Furthermore, based

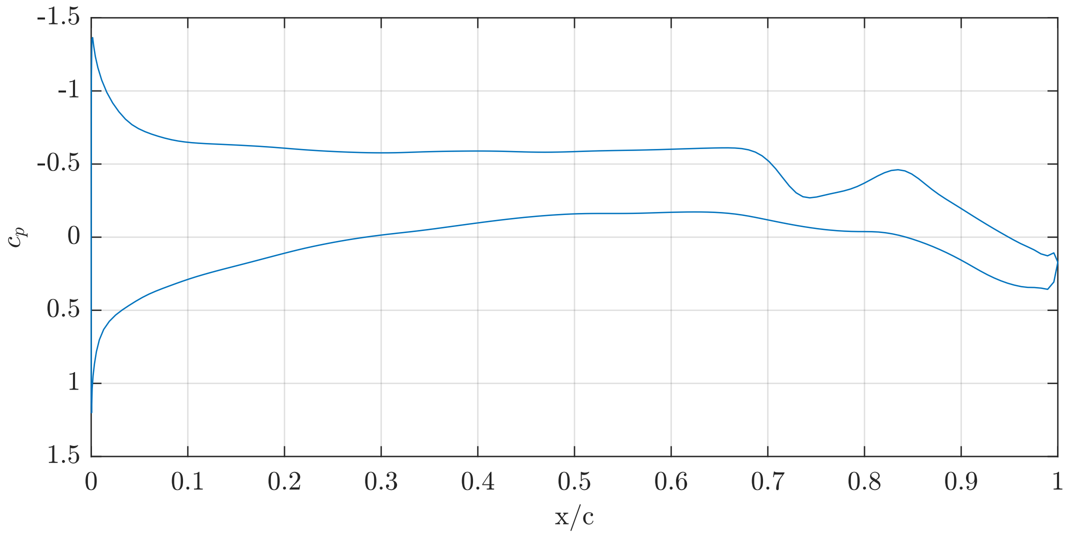

Figure 6: Optimized airfoil c_p distribution at the cruise operating condition.

on B factors provided in Nicolai, biconvex airfoils have a lower B-factor when compared to a hexagonal airfoil. This reduces wave drag that will occur during supersonic flight.

More investigation is required into the subsonic and supersonic performance of various sharp-nosed, symmetrical airfoils. This will ensure the vertical stabilizers, especially the all moving, horizontal stabilator, can provide sufficient control authority for the aircraft.

5 Fuselage Layout

5.1 Engine

The aircraft will utilize two F110-GE-132 turbofan engines. The F110-GE-132 has the highest thrust with lower dry TSFC and similar afterburner TSFC out of possible engines that were considered for the selection process. The F110-GE-132 has a max diameter of 46.5in, an inlet diameter of 35.66in, and total length of 181.9 in. The two F110-GE-132 engines are 28.0 in from the x-axis of the CAD to center of engine. The engines will be fed airflow through diverterless ducts configured with a forward swept cowl design, purposed to divert boundary-layer away from the aircraft's engine, and reduce radar reflections from the fuselage.

5.2 Avionics

5.2.1 AN/APG-81 Radar

The aircraft will utilize AN/APG-81 Active Electronically Scanned Array (AESA) radar. the AN/APG-81 AESA is a long range radar suitable for multi-role fighters and is currently used on the F-35 Lightning II. It has robust electronic warfare capabilities and is optimized for a multi-role mission requirement [7]. Since it is already in active use, it meet the technology readiness requirement set by the request for proposals (TRL 6 or higher). The radar is 2.3 feet in diameter and by using pixel counting of an image of the radar [8], it is determined that it is just over half a foot in length. This leads to a total volume of 2.8 ft^3. The weight of the radar is 485 lbs [9].

5.2.2 Other Avionics

According to Raymer, avionics volume can be estimated by using a density of 30-45 lb/ft^2. The total avionics weight is 2500 lbs (from the RFP), which fits into Raymer's estimates of avionics weight from aircraft empty weight (0.03 to 0.08 multiplied by the empty weight). Since the radar dimensions are known, that is subtracted from the weight of the rest of the avionics.

Since the general trend of avionics is a decrease in size [10], the higher end of the density range is chosen. This results in a volume of 44.8 ft^3 required for the rest of the avionics.

5.3 Weapons Bay

The weapons bay volume is constrained by the Air to Air Combat mission, where 6 AIM-120C's, and 2 AIM-9X's will be carried at the bottom of the fuselage similarly to the F22 configuration shown in Fig. 7 A weapons bay volume of 177 ft^3 will be required to carry this ordnance load. This will accommodate the 4 JDAM MK-83s for the air-to-ground mission. Volumes of independent ordnance objects were derived using their dimensions([11, 12, 13, 14]) to calculate the total volume required.

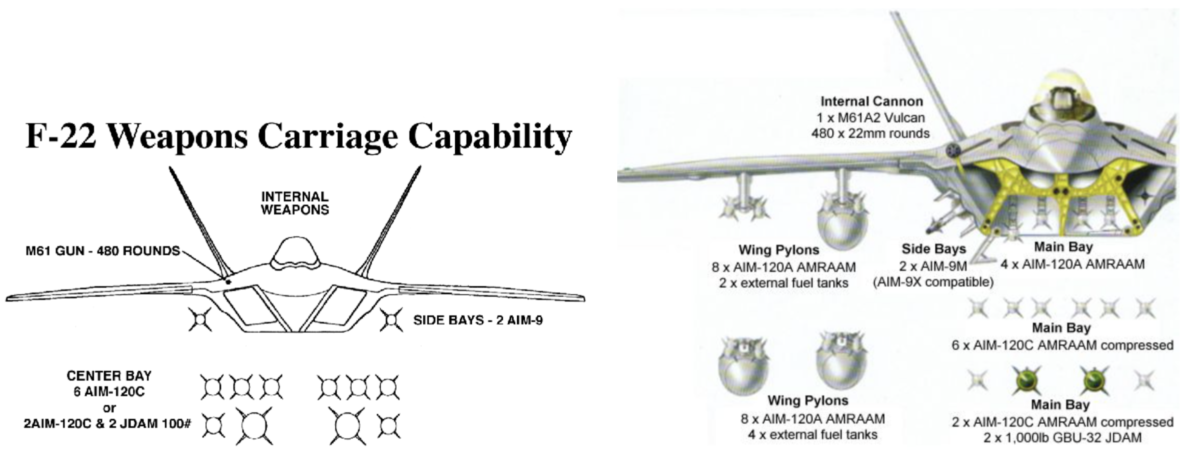

Figure 7: F22 Weapons Carriage Configuration Example. 6 AIM-120C's or 4 JDAM MK-83's will be carried in the main weapons bay with 2 AIM-9X's carried in side bays [15, 16]

.

5.4 Cockpit

Our cockpit will have a volume of 135 ft^3 with a length of 10.8 ft, width of 3.75 ft, and a maximum height of 3.33 ft. These values were derived from scaled two-dimensional drawings of the F-35C [17].

5.5 Spot Factor

The spot factor is 1.15, which is the design top-down stowed projected area divided by the F/A-18E's top-down stowed projected area. The F/A-18E's top-down stowed projected area was estimated by scaling by a top-down image by a known characteristic length and then estimating the top-down area with the wings folded using Siemens NX.

6 Fuel Tanks

The maximum amount of fuel needed for the required missions is 32,274 lbs. This is converted to gallons using the density of JP5 fuel: 6.74 lbs/gal [18]. 1 cubic foot is 7.48 gallons. The volume of fuel needed is 555.3 ft^3.

Fuel will be stored in external fuel tanks, the fuselage, the wings, and the vertical stabilizers, similar to the F-35 [17].

6.1 External Fuel Tanks

External fuel tanks will be located at 33.3% of the half span and at 12.5% of the chord fraction. Using Langrange [19], the center of gravity of the 480-gallon tank of the F/A-18 is in the middle of the tank. Looking at the F-22, F-16, and the Rafale,the center of gravity of the fuel tank is at 12.5% of the local chord. This selection also prevents the fuel tanks from interfering with the control surfaces. For the spanwise placement, Raymer guidelines for external store interference drag shows that interference drag, Q, is between 1.0-1.3 if it is located about a diameter away from the fuselage. Looking at the F-22, F-16, and Rafale, the fuel tank is located about a diameter away from the fuselage wall (a diameter and a half from the center of gravity of the fuel tank). Therefore, the center of gravity location of the fuel tank is about one and a half diameter away from the fuselage since it keeps the center of gravity of the fuel tanks close to the center of gravity of the fighter and lowers interference drag. This translates to a 33.3% half span location based on a fuselage width of 12 ft and the diameter of the 480-gallon fuel tank of 31.9 in [19]. There will be a 480-gallon tank on either side of the wing. This amount of fuel should get the aircraft to combat, and then they can be dropped.

6.2 Wing and Vertical Tail Tanks

The capacity of the vertical tail and wing fuel tanks are estimated using a trapezoidal prism along the surface, represented by Eq[1]:

$$V_{\text{fuel}} = \frac{a+b}{2}h \tag{1}$$

a and b are the chord lengths at the beginning of the surface (where the fuselage ends) up to the end of the usable surface. For the wing, the fuel will not be stored in the folding section to avoid over-complicating the system and increasing costs. Thus, each wing can only store fuel up to 17.5 feet, given the RFP requirement of a 35-foot wingspan with folded wings. So for the wing, b is 17.5 feet. For the vertical stabilizer, based on the configuration of the F-35 fuel tanks [17], b is the chord length at 90% of the vertical stabilizer length, estimated by an online pixel counter since exact dimensions could not be found [20]. Both a and b are multiplied by a factor, ch_{used}, to account for the part of the chord that can be used. In the F-35, that factor is 0.58 [17, 20]. The thickness is calculated as 4% of the average chord length over the length of the wing used (l_{used}) to account for the fact that not all the vertical space in the wing can be used for storage.

Thus, fuel volume is given by Eq.[2]:

$$V_{\text{fuel}} = (\frac{ch_{used}(a+b)}{2})^2 * l_{used} \tag{2}$$

The volume of fuel in the wings is 48 ft^3. The volume of fuel in the vertical tails is 31 ft^3.

6.3 Fuselage Tanks

The volume of fuel needed in the fuselage is given by Eq[3].

$$V_{\text{fuel}_{\text{fuselage}}} = V_{\text{fuel}_{\text{total}}} - V_{\text{fuel}_{\text{wings}}} - V_{\text{fuel}_{\text{tails}}} \tag{3}$$

The possible volume of fuel in the fuselage is calculated by assuming a rectangular prism around the location of potential fuel storage in the fuselage with the engines subtracted out. Based on the F-35 fuel tank layout [17], half the length of the fuselage is used. Once the engines are subtracted out, a factor of 0.5 is applied to account for half of the height of the fuselage being used. This height does not include the vertical stabilizer or landing gear. An additional factor is applied to account for the use-able space in this volume. Raymer applies a factor of 83% to account for use-able volume, so an additional factor of 0.8 is applied. The resulting equation is given in Eq.[4].

$$V_{\text{fuel}_{\text{fuselage}}} = 0.8(0.5(0.5 * l_{\text{fuselage}} * w_{\text{fuselage}} h_{\text{fuselage}} - n(V_{\text{engine}} + V_{\text{enginesupport}}))) \tag{4}$$

where n is the number of engines. The engine and engine support volume are calculated by assuming an engine diameter as large as the engine and the engine supporting structures put together. This provides a conservative estimate on fuel capacity and ensures that engine infrastructure is accounted for.

The needed fuel volume in the fuselage is 432.7 ft^3, which is less than the possible fuel volume, 508.0 ft^3.

6.4 F-35 Fuel Tanks

The fuselage and lifting surface dimensions used in the fuel tank calculations is shown graphically in Figure 8.

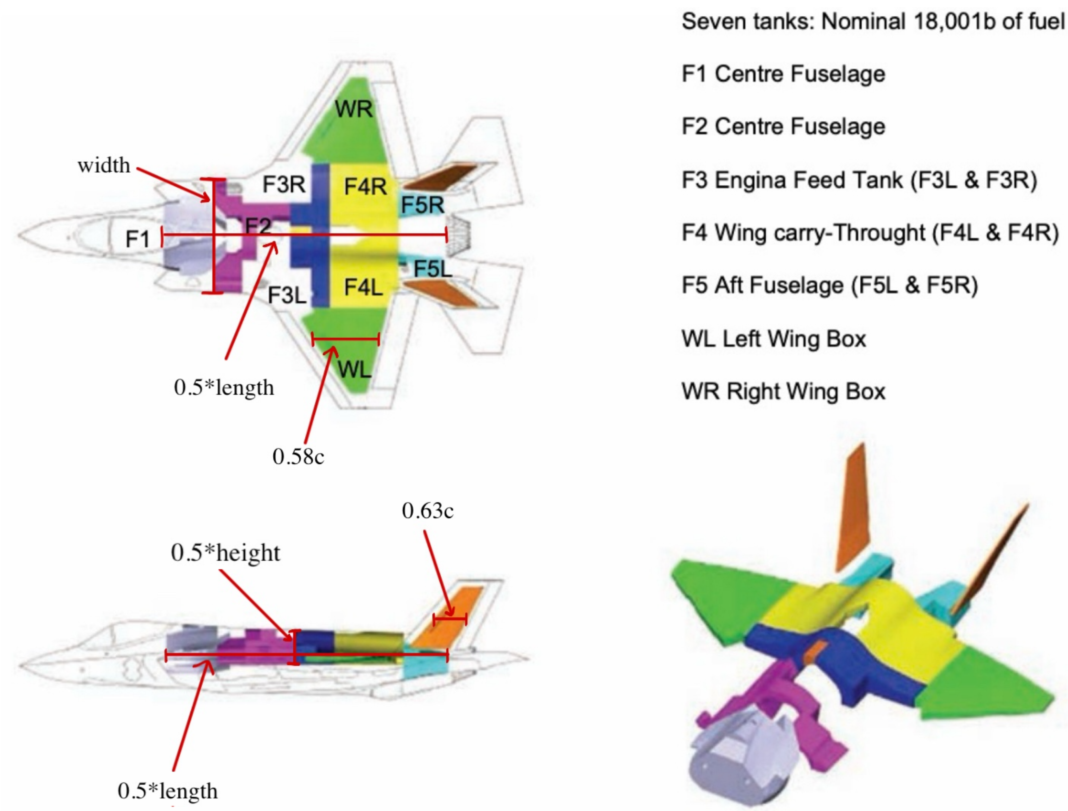

Figure 8: Reference Lengths for Fuel Estimate Calculations overlaid on the F-35 Fuel Tank layout [17]

7 Location of Key Components

The locations of the avionics, radar, fuel tanks, pilot seat, and weapons are estimated using the F-35 layout (see Fig. 8). The cockpit, radar, and avionics bay will be in the front third of the aircraft, the fuel tanks will be placed between 28.5% and 85% of the fuselage length, and the weapons are placed in an internal weapons bay starting behind the avionics bay and below the internal fuel locations. The external fuel tank location is covered in Section 6.1. The fuel tanks and avionics bay will not overlap to help prevent any potential safety issues.

Table 1: Location of Key Components in percentage of fuselage length

Start of Engines	62%
Cockpit	12-28%
Pilot Seat (x1)	20%
Avionics	12.5-28%
Radar	11%
Fuel Tanks	28.5-85%
Weapons Bay	38%-74%
External Fuel Tanks	50%

8 Group Work Split

Table 2: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	17%	Performed airfoil optimization and assisted with planform trade studies.
Charles Choi	17%	Created fuselage OML of aircraft, integrated wing configurations to fuselage OML, designed the fuselage inlets, helped engine/fuselage portion of report.
Cristina Erskine	17%	Researched internals/avionics and weapons bay, researched and calculated internal fuel volume, located key components.
James Gold	15%	Assisted with trade studies, researched historical tail sweep, found external fuel tank locations, found folded wing area of the F/A-18E.
Santiago Ramos-Assam	17%	Created parametrized CAD of the wings and empennage, calculated weapons bay and radar volume, researched historical tail sweep, helped write report.
Thomas Sheridan	17%	Created planform trade study tool using Excel. Fit drag polar data. Performed trade studies to optimize wing. Laid out horizontal and vertical tails. Sized control surfaces and flaps.

References

[1] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. ISBN: 9781624104909.

[2] *AVL 3.40*. 2022. URL: https://web.mit.edu/drela/Public/web/avl/avl_doc.txt.

[3] Joaquim R.R.A. Martins. *The Metabook of Aircraft Design*. Dr. Martins, Joaquim R.R.A., 2021.

[4] *Airfoil Optimization*. 2025. URL: https://mdolab-mach-aero.readthedocs-hosted.com/en/latest/machAeroTutorials/airfoilopt_overview.html#.

[5] *Standard Aircraft Characteristics F/A-18E Super Hornet*. NAVAIR00-110AF18-6. Boeing. 2001.

[6] Leland M. Nicolai. *Fundamentals of Aircraft Design*. American Institute of Aeronautics and Astronautics, 2010. ISBN: 978-1-60086-751-4.

[7] *AN/APG-81 Active Electronically Scanned Array (AESA)*. 2025. URL: https://www.northropgrumman.com/what-we-do/mission-solutions/radars/an-apg-81-active-electronically-scanned-array-aesa-fire-control.

[8] *AN/APG-81*. 2016. URL: https://www.deagel.com/Components/ANAPG-81/a001381.

[9] *F-35's Advanced AN/APG-81 Radar*. 2010. URL: https://www.scribd.com/document/29267055/ANAPG81#:~:text=Weight:%20220%20kg%20%28reported%29,laboratory..

[10] Adam J. Fleischer. *Miniaturized Electronics Propelling the Future of Aerospace*. 2023. URL: https://www.engineering.com/miniaturized-electronics-propelling-the-future-of-aerospace/#:~:text=One%20of%20the%20remarkable%20advancements,and%20efficiency%20are%20significantly%20enhanced..

[11] *AIM-120 Advanced Medium-Range, Air-to-Air Missile (AMRAAM)*. URL: https://www.navy.mil/Resources/Fact-Files/Display-FactFiles/Article/2168352/aim-120-advanced-medium-range-air-to-air-missile-amraam/ (visited on 09/10/2025).

[12] *AIM-9X Sidewinder Missile*. URL: https://www.navy.mil/Resources/Fact-Files/Display-FactFiles/Article/2168989/aim-9x-sidewinder-missile/ (visited on 09/10/2025).

[13] *Joint Direct Attack Munition (JDAM)*. URL: https://www.navy.mil/Resources/Fact-Files/Display-FactFiles/Article/2166820/joint-direct-attack-munition-jdam/ (visited on 09/10/2025).

[14] *Joint Direct Attack Munition GBU- 31/32/38*. URL: https://www.af.mil/About-Us/Fact-Sheets/Display/Article/104572/joint-direct-attack-munition-gbu-313238/ (visited on 09/10/2025).

[15] Jeff Scott. 2002. URL: https://aerospaceweb.org/question/planes/q0105.shtml.

[16] URL: https://www.globalsecurity.org/jhtml/jframe.html#https://www.globalsecurity.org/military/systems/aircraft/images/f-22-weaps-mg27.gif%7C%7C%7C.

[17] Allan Seabridge Ian Moir. *Aircraft Systems: Mechanical, Electrical, and Avionics Subsystems Integration, Third Edition*. John Wiley and Sons Ltd, 2008. ISBN: 978-1-56347-952-6.

[18] *Standard Practice Quality Assurance/Surveillance for Fuels, Lubricants and Related Products*. 2016. URL: https://www.dla.mil/Portals/104/Documents/Energy/Quality%20and%20Technical%20Support/E_MilSTD3004_1603.pdf (visited on 09/14/2025).

[19] Mario B.J. Lagrange. "Flight Test Certification of a 480-Gallon External Fuel Tank on CF-18". In: *Proceedings 8th JOCG Aircraft/Stores Compatibility Symposium*. 1990.

[20] *Online Pixel Ruler*. URL: https://www.rapidtables.com/web/tools/pixel-ruler.html.